

HUMAN ORIENTATION DISCRIMINATION TESTED WITH LONG STIMULI

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Abstract—Human orientation discrimination was measured for single long lines presented sequentially. Orientation sensitivity is better for a narrow range of orientations around the principal meridians. Both the orientation sensitivity and its meridional variation increase with stimulus length. Control experiments show that the "oblique effect" can be obtained as well with a signal detection method as with the method of constant stimuli and the subjects used a visual orientation cue in their judgements. Comparison of these findings with properties of visual cortical cells suggest that cells similar to S cells of cat and monkey striate cortex, carry the neuronal representation used in the decision process. And indeed the meridional variation in orientation sensitivity can be predicted from the number of monkey S cells preferring different orientations.

INTRODUCTION

Apart from absolute line orientation estimation which is a method of questionable validity (Matin and Drivas, 1979), most parametric studies on human orientation discrimination used matching procedures. In these studies subjects has to set a line parallel to another one (e.g. Rochlin, 1955; Andrews, 1967) or to place a dot in extension of a line (Bouma and Andriessen, 1968). In such tasks configurational effects and vernier like mechanisms may be involved. Moreover, in all existing relevant electrophysiological work on orientation selectivity of visual cortical cells only a single line (or grating) is used as stimulus. Hence, correlation of the results of the above mentioned studies with physiology is difficult. In this respect, a successive discrimination task in which single lines are presented in a frameless environment, would be a more appropriate test of orientation sensitivity.

Previous studies on human orientation discrimination used rather short lines as stimuli, e.g. Matin and Drivas (1979) used lines of 0.25° . Electrophysiological studies on cat (Kato *et al.*, 1978) and monkey (Schiller *et al.*, 1976) showed that the optimal length is 1.4° for end-stopped visual cortical cells and 6.5° for end-free cells. The latter value is considerably larger than the lengths used in previous studies on human orientation discrimination.

It has been suggested that orientation sensitivity is better around principal meridians than around oblique ones (Andrews, 1967; Bouma and Andriessen, 1968). Recent evidence strongly suggests that both in cat and monkey more cells of a subclass of area 17 cells (at least with a receptive field less than 10° from fixation point) prefer orientations close to

the principal meridians than orientations close to the oblique ones (Berman *et al.*, 1981; Pettigrew *et al.*, 1968; Leventhal and Hirsch, 1980; Orban and Kennedy, 1981). This subclass, the S cells in the terminology of Orban and Kennedy (1981), have narrow receptive fields with single or separate On and Off subregions and show considerable length summation. Since S cells are the only receptive field type of area 17 to show a meridional variation in orientation preference (Table 1) and are as a group the most narrowly tuned for orientation, it is likely that these S cells are involved in orientation discrimination. From this hypothesis, two predictions can be derived. First, for long stimuli orientation discrimination should be better for orientations within 15° from a principal meridian and equally worse outside this range. This follows from the distribution of preferred orientations of this subclass of area 17 cells (Table 1). Second, due to the considerable length summation of S cells, the meridional variations in orientation discrimination should be much stronger for long than for short lines. We have tested these predictions in an experimental set up in which the stimuli were presented in a way similar to that used in neurophysiological studies. We have presented in succession single lines of different orientations centered around a reference orientation. The stimuli were viewed monocularly in a frameless environment. Line length was varied from 0.5 to 18° visual angle. Other independent variables studied were reference orientation, experimental design and body orientation.

METHODS

Stimuli

The stimulus was a fluorescent rod attached to the rear panel of a metal box, rotating about its center,

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Table 1. (A) Median ORT tuning width

		S family	B family	C family	A family
<i>Cat</i> ^c					
Area 17	Free ^a	44(51)	61(6)	80(33)	65(10)
(V1)	H	74(38)	78(2)	73(12)	84(4)
Area 18	Free	58(61)	69(15)	88(28)	98(16)
(V2)	H	76(14)	63(5)	91(8)	125(2)
Area 19	Free	65(7)	55(4)	107(18)	85(7)
(V3)	H	80(13)	110(11)	92(17)	82(8)
<i>Baboon</i> ^{d,f}					
	V1	41(59)	54(29)	92(22)	57(8)
	V2	42(21) ^e	52(17)	58(30)	48(10)

(B) Number of V1 cells ($\text{ecc} < 10^\circ$) having a preferred orientation of a given obliquity^d

S cells	0-9	10-18	19-27	28-36	37-45
<i>Cat</i> ^c	22	22	5	10	15
<i>Baboon</i> ^f	18	9	5	5	5
Total population					
<i>Cat</i> ^c	86	82	43	43	51
<i>Baboon</i> ^f	34	33	18	21	25

^aFree: end-free cells, H: end-stopped cells.^bSince there was no difference between them, data are pooled for end-free and end-stopped number of the families.^cOccur only in a narrow strip adjacent to the V1-V2 border.^dObliquity: orientation relative to a principal meridian (H or V). Numbers between brackets indicate the number of cells of a given type for which orientation tuning was measured.^eAccording to Orban.^fAccording to Kennedy, Martin, Orban and Whitteridge, in preparation. These cells were recorded in 16 oblique penetrations made into area 17 of 5 baboons.

and illuminated by u.v. light. On the front panel, openings closed by shutters, allowed the subject to view the rod inside the box either with his left or right eye. Since the inner walls of the box are painted black the rod appeared in an empty dark field. The rod fluoresced red and had a luminance of 0.34 cd/m^2 while the background illumination was 0.15 cd/m^2 . Consequently the contrast ($\log I/I$) of the stimulus was 0.1. At the viewing distance used (57 cm) the rod was $14'$ min arc wide and unless specified otherwise 18° long. The orientation could be adjusted manually in steps of $1/4^\circ$. The presentation time was controlled by means of the shutter and fixed at 600 msec in all experiments. The subject had to respond within this time interval. The intertrial interval was about 5 sec.

Subjects

Subjects were both male and female students ranging in age from 20 to 25 years. A third of them were naive about the experiment, others not. All had normal vision or corrected spherical refraction errors. None had an astigmatism larger than 0.50 D. In pilot studies no differences were seen whether the subjects used the dominant or non-dominant eye and in the main study subjects always used their right eye.

Method of constant stimuli

With one exception, the method of constant stimuli was used to determine the differential threshold or

JND in orientation at a given reference orientation. This determination was done in two steps: a training and the determination itself. During the training the subject had to discriminate between 2 orientations, symmetrical with respect to the reference orientation. The difference between both orientations was gradually decreased until the performance of the subject fell below 90% correct responses. The smallest orientation difference for which the subject reached 90% fixed the interval used in the JND determination. Five orientations dividing the interval into 4 equal parts were used for determination of the JND. The subject had to press one key for orientations tilted clockwise from the reference orientation and the other for orientations tilted anticlockwise. Each orientation was presented 64 times in random order without feedback and each differential threshold is based on 320 responses. The proportion of pressing a key as a function of stimulus orientation approximately follows a cumulative normal distribution. By probit analysis the mean and standard deviation (SD) of the distribution could be estimated. The SD was taken as the JND in orientation.

Signal detection paradigm

In this paradigm we followed the general methods of Marin and Drivas (1979). In this method a pair of orientations, symmetrical about the reference orientation, were presented sequentially 10 times each, in

Table 2. Stability of JNDs measured at the four main orientations in our subject (R.V.)

	H	V	LO	RO	OEI
1st testing	1.06	—	4.31	—	307
2nd testing	0.93	0.86	1.79	2.23	126
3rd testing	0.69	0.71	2.56	1.81	211
4th testing	0.57	0.82	1.45	1.78	131
5th testing	0.76	0.96	2.71	2.81	221
Mean (last 4 tests)	0.74	0.84	2.13	2.16	172
SD	0.15	0.1	0.61	0.48	51

random order and with feedback. A measurement consisted of a series of 7 blocks, each block corresponding to a different orientation difference. For each experimental condition seven measurements were made and the order of the blocks within the measurements were randomized according to a latin square design. The slope of the plot of d' as a function of orientation difference is a measure of orientation sensitivity (provided the lines pass the origin).

RESULTS

Orientation sensitivity as a function of orientation of long lines

In experiment 1, JNDs in orientation were measured with a 18° long line at the 4 main orientations: horizontal, vertical and left and right oblique. The order of testing was randomized and different for each subject. The magnitude of meridional variation is assessed by computing an oblique effect index

$$(OEI) = \frac{JND_o - JND_p}{JND_p} \times 100$$

where JND_o is the mean JND in orientation measured at oblique orientations and JND_p the mean JND in orientation measured at principal orientations (horizontal and vertical). One of the authors (R.V.) was a subject in most of the experiments and JNDs at the main 4 orientations were measured in different occasions spanning more than 1 year. Except for a decrease in threshold between the first and second testing the thresholds are stable and the OEI oscillates between 125 and 225 (Table 2). Because of this initial improvement in the oblique JNDs, we accept among our reference population of normal young adults only those which had 2 training JND measurements before measuring the JNDs in the 4 main orientations. The data of 9 subjects fulfilling these requirements are shown in Fig. 1. There was no difference between naive and non-naive subjects.

JNDs in orientation measured at principal orientations (horizontal and vertical) are much smaller than those measured at oblique orientations (Fig. 1). As indicated by the OEI (mean 183, SD 66) the JNDs

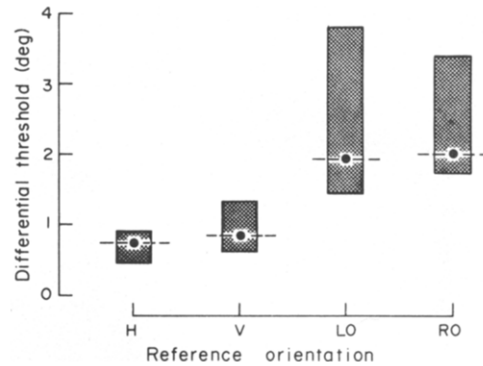


Fig. 1. Distribution of JNDs in orientation at the four main reference orientations for 9 subjects. Both the medians (dots) and range (hatching) are shown. H: horizontal, V: vertical, LO: left oblique, RO: right oblique.

at oblique orientations are 2 to 4 times larger than JND at principal orientations. Both JNDs at horizontal and vertical differ significantly ($P < 0.01$) from the JNDs at oblique orientations. In fact none of the distributions overlap as indicated by the $T = 0$ value in the Wilcoxon matched pairs signed ranks test. While JNDs at both oblique orientations do not differ, there is a small, just significant ($P < 0.05$) difference between JNDs at horizontal and vertical.

In the second experiment JNDs were measured at 15 orientations symmetrical around left oblique and ranging between horizontal and vertical. Order of testing was randomized. For two subjects (L.B. and F.V.C.) complete data were obtained, for a third subject (P.S.) mainly half of the curve, between horizontal and left oblique was obtained.

Figure 2 shows the JNDs plotted as a function of orientation for subjects L.B. and F.V.C. Both curves are symmetric. Given this symmetry and similarity in JNDs for horizontal and vertical, the data have been replotted as a function of obliquity, i.e. orientation difference with respect to a principal orientation (horizontal and vertical). Given the symmetry in the curves, data of subject P.S. could be pooled with the others. The mean JND as a function of obliquity for the 3 subjects is plotted in Fig. 3(A). JND in orientation increases as a function of obliquity from the principal orientation up to 20° obliquity and then

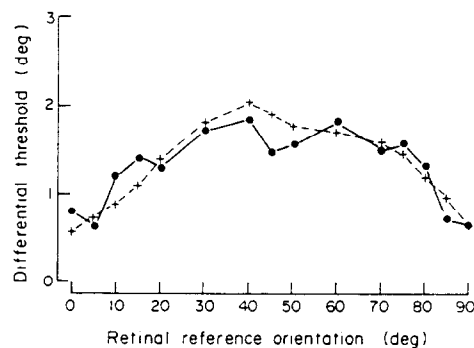


Fig. 2. JNDs in orientations plotted as a function of retinal reference orientation for subjects F.V.C. (dots) and L.B. (crosses).

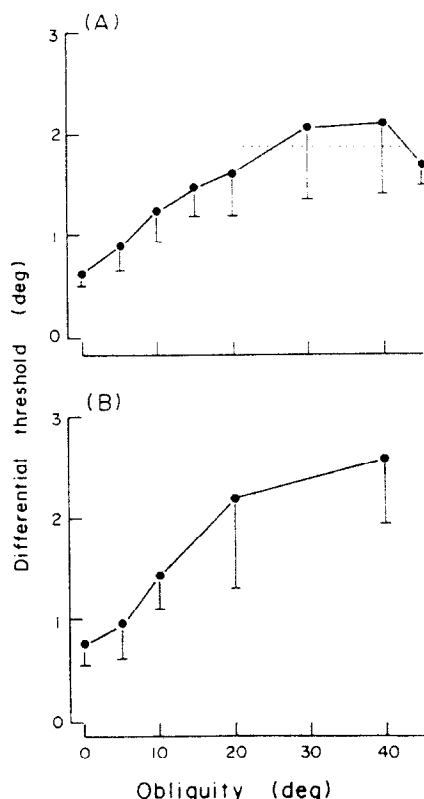


Fig. 3. JNDs in orientations plotted as a function of obliquity (orientation difference with a principal meridian). (A) Mean and standard deviation for 5 measurements in 3 subjects. Stippled line: mean JND for obliquities over 20°. (B) mean and standard deviation for 13 measurements in 5 subjects.

levels off. This experiment confirms the prediction that over a broad range of orientations ($\pm 50^\circ$) around the oblique orientations discrimination is equally weak.

Differential orientation sensitivity for different line lengths

Experiment 1 demonstrates a strong consistent oblique effect in orientation acuity for long lines. Matin and Drivas (1979) performed a similar discrimination experiment but with short lines ($1/4^\circ$) and

also reported an oblique effect in orientation acuity. Our hypothesis predict a stronger oblique effect in our experimental conditions than in those of Matin and Drivas (1979) but the experiments are not immediately comparable since they also differed in stimulus width and luminance and in the psychophysical method. Therefore we have compared in a third experiment, the orientation discrimination at four main orientations for short (0.5°) and long (18°) stimuli with two psychophysical methods (constant stimuli and signal detection). The 8 JNDs and 56 measurements for the signal detection method were divided into 64 sessions which were randomized. Two subjects (R.V. and A.P.) participated in the study.

In the signal detection method the sensitivity is indicated by the slope of the psychometric functions. The inverse of the slope is a measure which can be compared to thresholds. These inverses are indicated for the 8 conditions in Table 3 along with the 8 JNDs.

It is plain that with both methods long stimuli give a strong oblique effect in orientation acuity while short stimuli induce a weaker effect. Since only 2 subjects and 4 orientations were tested, only 8 datapoints were available for analysis of variance. Given this small number of datapoints a general linear model of analysis of variance (Timm, 1975) was used. This method uses the datapoints on which the slopes of signal detection or the JNDs are calculated rather than the slopes or the JNDs themselves. For the slopes of the signal detection method, the lambda values for comparison of principal and oblique orientations were very significant for long stimuli, and not or weakly significant for short stimuli (Table 3). Direct testing for the effect of line length onto the orientation effect yields a very significant lambda value for both subjects. A similar analysis performed on the 8 JNDs gives analogous results, although for short stimuli the comparison of principal and oblique orientations also yielded significant lambda values. The variability of the "oblique effect" for short line lengths is further illustrated by the observation that for short stimuli the subject who had the largest effect with the constant stimuli was not significantly influenced by the reference orientation in his signal

Table 3. Meridional variation in orientation acuity for long and short stimuli, assessed with method of constant stimuli and with signal detection method

(A) Signal detection (slope ⁻¹)											
Subjects	Long lines orientations					Short lines orientations					Long vs short λ
	H	V	LO	RO	λ	H	V	LO	RO	λ	
R.V.	1.0	1.2	2.5	2.65	0.49 ^b	4.8	3.3	4.2	7.1	0.93	0.58 ^b
A.P.	1.8	1.3	3.9	4.6	0.34 ^b	7.1	5.6	7.7	16.7	0.77 ^a	0.5 ^b
(B) Method of constant stimuli (JNDs)											
Subjects	Long lines orientations					Short lines orientations					Long vs short λ
	H	V	LO	RO	λ	H	V	LO	RO	λ	
R.V.	0.69	0.71	2.56	1.81	0.27 ^b	2.46	2.46	4.19	5.78	0.44 ^b	0.39 ^b
A.P.	0.63	1.12	3.79	2.8	0.12 ^b	3.42	4.58	5.31	6.87	0.71 ^b	0.16 ^b

^a $P < 0.05$.

^b $P < 0.01$.

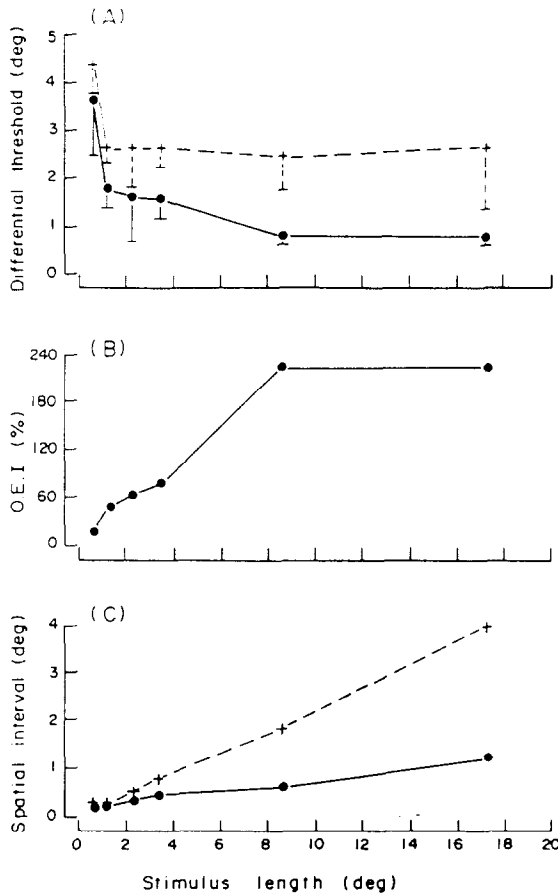


Fig. 4. (A) JNDs in orientations plotted as a function of stimulus length. The mean and standard deviation for 4 subjects are indicated. (B) Mean (4 subjects) oblique effect index plotted as a function of stimulus length. (C) Mean (4 subjects) spatial interval between endpoints corresponding to JND plotted as a function of stimulus length for principal orientations (continuous line) and for oblique orientations (dashed lines). Note the initial flat part of the curve.

detection performance. These experiments demonstrate that the orientation acuity oblique effect is stronger and more consistent for long than for short stimuli, independently of the psychophysical method used.

In the previous experiment only two stimulus lengths were used. Experiment 4 was undertaken to study orientation discrimination and its meridional variations as a function of stimulus length. Two orientations (horizontal and left oblique) and 6 lengths were tested in a randomized design. Four subjects (R.V., L.M., L.B. and F.V.C.) participated in this experiment. Figure 4(A) shows the mean JND in orientation plotted as a function of stimulus length for horizontal and oblique orientations. For both orientations JNDs decrease in parallel up to 1° length. Over 1° length discrimination does not improve for the oblique orientation while for the horizontal orientation it improves up to 8° length. Consequently the OEI increases up to 8° length and then saturates [Fig. 4(B)]. As predicted the oblique effect increases with increasing line length.

Position cues vs orientation cues (experiment 5)

While the previous experiment clearly shows the predicted effect of stimulus length, it could simply be due to the use of the position of the end-points rather than orientation of the line as cue. Indeed the difference in position of the ends of two lines of length l , differing in orientation by an angle x (smaller than 5°) equals $(l \sin x)/2$. If the subject used the position cue rather than the orientation cue one would predict an improvement with length and the meridional variations could then be the reflection of the "ordinary" oblique effect in acuity. If this were true then for the different lengths, $[l \sin (JND)]/2$ should yield approximately the same value. The results of the previous experiment (experiment 4) have been replotted in that way in Fig. 4(C). Both for the oblique and horizontal orientation $[l \sin (JND)]/2$ is not independent of line length and on the contrary, increases in a systematic way.

Given the importance of this issue a separate experiment (experiment 5) was conducted in which orientation and position discrimination were compared. JNDs were measured for the 18° long line and for a dot (15 min arc diameter) placed at the end of a (invisible) rod of the same length. Since the change in position of the end-point of the line and of the dot are equal, the experiment compares the contribution of position and orientation cues to the discrimination task. Although there was no control of fixation, the subjects very likely fixated the dot for the discrimination since the dot was the only visible object in a frameless environment. Therefore both stimuli did differ in eccentricity as little as possible. The order of testing of the 8 conditions (4 orientations and 2 stimuli) were randomized. Two subjects (R.V. and M.D.W.) participated in this experiment. Figure 5 gives the results of the experiment. For the long lines the JNDs in orientation are plotted and for the dots the change in orientation of the carrier rod corresponding to the position threshold. The position discrimination is much worse than orientation dis-

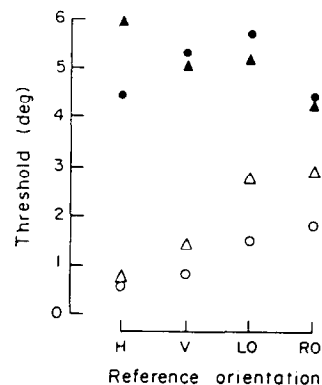


Fig. 5. JNDs in orientations for long lines (open symbols) and equivalent differential thresholds for dots placed at the end of an invisible rod of same length (solid symbols) at the 4 main orientations (of the line or invisible rod). Abbreviations as in Fig. 1. Circles: subject R.V., triangles: subject M.D.W.

crimination and more importantly shows no meridional variation. The OEI calculated on the thresholds for the dots are 5 and -13 for subject R.V. and M.D.W. respectively. Both Fig. 4(C) and this experiment show that the subjects used more than just position information in the orientation discrimination with long lines. The initial flat part of the curve in Fig. 4(C), suggests that for short stimuli (less than 1°) subjects may use position cues in orientation discrimination tasks.

Do the meridional variations in orientation discrimination follow retinal or gravitational coordinates (experiment 6)

The previous experiments (experiments 1 and 2) show that JNDs in orientation are minimum around horizontal or vertical. If this meridional variation does not depend on vestibular references influencing the decision it should follow retinal coordinates and when the subjects are tilted the minimum in the JNDs in orientation-orientation curve should shift almost exactly the amount of body tilt. The small difference between the degree of tilt and shift will be due to the counter-rolling of the eye. The subjects' body was tilted 20° (clockwise for subject R.V. and anti-clockwise for subject M.D.W.). The counterrolling of the eyes under these conditions, is about 3° (Merker and Held, 1981; Miller, 1962). Nine orientations centered around a retinal principal orientation were tested both in tilted and untilted conditions. These 18 conditions were randomized. This test was done for long lines (15°) both around vertical and horizontal and for short lines (0.5°) around vertical.

The data for both principal orientations of long stimuli are pooled for each subject. The data of both subjects for short stimuli are also pooled. They are shown in Fig. 6 in which JNDs in orientations are plotted as a function of orientation in gravitational coordinates. The curves for the untilted conditions confirm all the previous experiments (2, 3 and 4): only for long stimuli and for orientations close to a principal orientation is orientation discrimination clearly better. In fact the curves for long stimuli supplement the data of the previous experiment (2). In order to improve our estimate of changes in JND in orientation as a function of obliquity they were pooled with the results of experiment 2. The outcome is shown in Fig. 3(B) in which each mean is at least calculated from 10 measurements. The Wilcoxon signed rank test yields very significant ($P < 0.01$) differences in JND between 40° obliquity on one hand and 0° , 5° and 10° obliquity on the other, but not between JNDs at 40° and 20° obliquity. This demonstrates that orientation sensitivity is significantly greater over a narrow (20°) range around the principal orientations and weaker over a broad (50°) range around the obliques. For the short lines the JNDs do not change in a systematic fashion as a function of orientation confirming experiment 3 and 4.

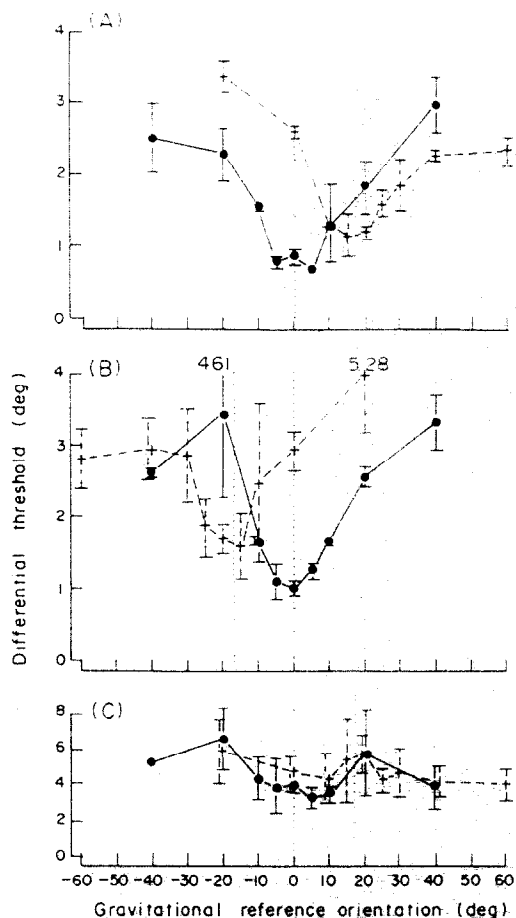


Fig. 6. (A) JNDs in orientations plotted as a function of gravitational reference orientation for subject R.V. in untilted position (continuous line) and tilted 20° clockwise position (stippled lines). The stimulus was a 15° long line. Zero represents a principal meridian (horizontal or vertical) and negative orientations represent orientations counter-clockwise from principal meridian. The JNDs for orientations around both principals and their mean are plotted. (B) Similar data for subject M.D.W. in untilted and 20° counter-clockwise tilted position. (C) JNDs in orientations plotted as a function of gravitational reference orientation for short lines (0.5°) in untilted (continuous line) and 20° clockwise tilted position (stippled line). Only orientations around vertical were tested. The values for both subjects (R.V. and M.D.W.) and their mean are shown. Stippled vertical lines represent retinal principal orientations assuming a 3° counterrolling of the eyes for a 20° body tilt (Merker and Held, 1981; Miller, 1962).

For long stimuli tilting the body induces three changes in the JND in orientation-orientation curves (Fig. 6): (1) it shifts the minimum over about 20° (2) it raises slightly the minimum of curve, i.e. JNDs around the retinal principal orientations and (3) it makes the curve asymmetrical: JNDs at the end towards which the body is tilted decrease and those at the opposite end increase. The shift of the minimum in the curve makes it plain that the meridional variation in JNDs in orientation follows retinal and not gravitational coordinates. The curves for short stimuli [Fig. 6(C)] show little difference between untilted and tilted conditions.

DISCUSSION

Our control experiments (5 and 6) show that when presented with long stimuli the subjects use mainly if not exclusively a visual cue in their discrimination task and that this cue is the retinal orientation of the stimulus. The contribution of possible vestibular afferents to the decision process is therefore small or negligible. Our discrimination task with long lines may be a more valid test of orientation sensitivity than previous experiments supposedly doing so (Andrews, 1967; Bouma and Andriessen, 1968). Configurational effects are unlikely to occur in our set-up and except for the short line lengths the subjects did not use position cues. The latter observation confirms the results of Westheimer (1981), although our thresholds are large compared to those of Westheimer. This may be due to the low level of background illumination or the absence of visual references in our experiments.

The successive presentation of different orientations may increase the role of the memory store. It should however be noted that even in a discrimination task in which 2 stimuli are present simultaneously, a memory "groove" may be important (Johnson, 1980; Matin and Drivas, 1979). The observation that the meridional variations in orientation discrimination can be demonstrated with different psychophysical methods, makes it unlikely that the origin of these variations resides in the decision process. This leaves the neuronal representation of the stimulus (Johnson, 1980) or the memory store as source of the meridional variations. The effect of line length renders the latter possibility unlikely. Hence, the meridional variations in orientation sensitivity in all likelihood are due to the properties of the cells supporting the neuronal representation of visual orientation. The fact that both predictions made in the introduction are verified, strengthens but does not prove our hypothesis that cells similar to V1 S cells of cat and monkey, are involved in orientation discrimination for long lines. The improvement of orientation discrimination with line length could be due to length summation and/or recruitment of additional cells of any type. Since the improvement for lengths over 1° is limited to the principal orientations and since only S cells in V1 show meridional variations in orientation preference (Orban and Kennedy, 1981, Table 1), it is difficult to see how other cell types as e.g. complex cells could contribute to orientation discrimination of long lines. It should also be stressed that physiological results (Bullier and Henry, 1979; Bullier and Henry, 1980) strongly suggest that the different receptive field types are separate functional streams operating in parallel within V1. This cortical parallel processing could indicate that the different streams (as e.g. the S cells) are specialized in the representation of different aspects of the visual scene (as e.g. orientations).

Recently Vandenbussche and Orban showed that the orientation discrimination of the cat is better

around principal orientations than around oblique orientations (Vandenbussche and Orban, 1983) and that this oblique effect also increases with line length (Vandenbussche *et al.*, 1983). This suggests that the cat is a reasonably good animal model for human orientation discrimination. Hence it could well be that in the two species similar cell types carry the neuronal representation of stimulus orientation.

Both in cat and monkey meridional variations involve mainly the orientation preference of S cells and hardly their width of tuning. This may indicate that the meridional variations in orientation discrimination depend more on changes in number of S cells active, than on a change in their width of tuning. There is a good fit between the actual changes in human orientation discrimination and those expected from the number of S cells in the baboon. According to sampling theorem, the signal to noise ratio increases proportionally to the square root of the number of active S cells. The number of S cells preferring a given obliquity, as observed by Kennedy, Martin, Orban and Whitteridge (unpublished) is shown in Fig. 7(A). The expected ratio of sensitivity is given by the square root of the ratio: number of S cells for a given obliquity/mean number of S cells for obliquities between 20° and 45° . The observed ratios of sensitivity are derived from Fig. 3(A) and are given by the ratio of sensitivity at a given obliquity and the mean sensitivity for obliquities between 20° and 45° . The fit [Fig. 7(B)] is excellent. This good fit suggests

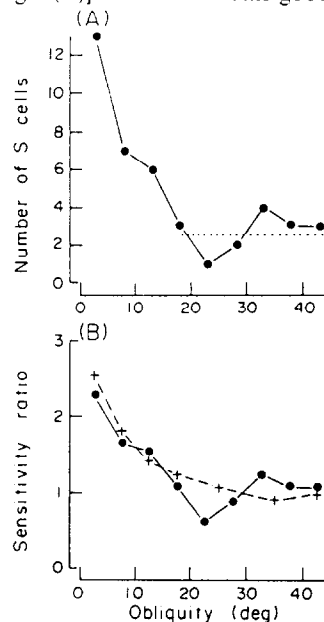


Fig. 7. (A) Number of S cells (see Introduction for definition) of the baboon plotted as a function of obliquity, according to Kennedy, Martin, Orban and Whitteridge (unpublished). Stippled line: mean number of S cells for obliquities over 20° . (B) Actually observed (stippled line) and predicted (continuous line) sensitivity ratio for a given obliquity and obliquities over 20° . The actual ratio is the ratio between JNDs in Fig. 3(A) and the mean JND for obliquities over 20° . The predicted ratio is the square root of the ratio of the number of S cells at a given obliquity and the mean number of S cells for obliquities over 20° .

that for long stimuli orientation discrimination performance is set by the number of cells active at this level of neuronal representation of stimulus orientation. The meridional variation of orientation preferences of the overall V1 cell population (Mansfield, 1974; Blakemore *et al.*, 1981; De Valois *et al.*, 1982; Table 1) is by far too small to explain the variations in orientation discrimination. This further supports the idea that only a subgroup of V1 cells carries the neuronal representation used by the decision process in human orientation discrimination of long lines.

Experiment 6 shows that the meridional variations in differential orientation sensitivity, as those in acuity and vernier acuity (Lennie, 1974; Banks and Stolarz, 1975; Corwin *et al.*, 1977) depend on retinal rather than gravitational coordinates. It is thus likely that the meridional variations in orientation acuity have their physiological basis in the properties of visual cells. This together with our other experimental observations points towards V1 as source of the meridional variation in orientation acuity, as has been suggested for the oblique effect in acuity. Essock (1980) suggested that the oblique effect in acuity could depend on the X receiving visual cortical cells. The populations underlying both types of meridional variations are probably different since S cells receive both X and Y afferents (Bullier and Henry, 1979; Bullier *et al.*, 1982). This may explain why behaviorally cats show an oblique effect in orientation discrimination but not in acuity (Bisti and Maffei, 1974; Vandenbussche and Orban, 1983).

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